There are six problems in total. You must solve the first five, and problem 6 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^n 1^m : n = 4q + r_1, m = 4q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 3\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ is a palindrome}\}$$
$$L_2 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : \text{every second letter in } w \text{ is } \mathtt{a}\}$$
$$L_3 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ contains more b than a}\}$$
$$L_4 = L_1 \cap L_2$$
$$L_5 = L_2 \cap L_3$$

Now solve the following problems.

(a) **Give** a context-free grammar that generates $\overline{L}_1$. (3 points) [Recall: $\overline{L}_1$ denotes the complement of the language L_1 i.e., $\overline{L}_1 = \Sigma^* - L_1$]

(b) **Give** a context-free grammar that generates L_2. (3 points)

(c) **Give** a context-free grammar that generates L_3. (3 points)

(d) **Write** all five-letter strings in L_4. (1 point)

(e) **Give** a context-free grammar that generates L_4. (2 points)

(f) **Give** a context-free grammar that generates L_5. (2 points)

(g) **Find** the total number of strings that belong to $L_5 \setminus L_1$. (1 point) [Recall: $L_5 \setminus L_1$ contains all strings that are in L_5 but not in L_1.]

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \to P0Q$$
$$P \to 0P0 \mid 0P1 \mid 1P0 \mid 1P1 \mid 0 \mid 1$$
$$Q \to 0Q1 \mid 1Q0 \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string '0110101010'. (3 points)

(b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)

(d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ starts with } 1 \text{ and ends with } 0\}$$

$$L_2 = \{w = 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } i = j + 2k\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and the number of 1s in } w_1 \text{ is equal to the number of 0s in } w_2\}$$

Now solve the following problems.

(a) **Give** the state diagram of a PDA that recognizes L_1. (3 points)

(b) **Give** the state diagram of a PDA that recognizes L_2. (3 points)

(c) **Give** the state diagram of a PDA that recognizes L_3. (4 points)

Problem 5 (CO4): Turing Machines (10 points)

Consider the following languages.

$$L_1 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and } \text{count}(w_1, 1) = \text{count}(w_2, 1)\}$$

$$L_2 = \left\{w \# w^R : w \in \{0,1\}^*\right\}$$

(a) For the configuration $10q_3 1011$ of a Turing Machine, **what** is the current position of the tape head? (1 point)

(b) **Give** the state diagram of a turing machine that decides L_1. [Note: $\text{count}(w, a)$ denotes the number of occurrences of the symbol a in the string w.] (3 points)

(c) **Give** the state diagram of a turing machine that decides L_2. [Note: w^R denotes the reverse of w, for example, if $w = 01101$, then $w^R = 10110$.] (4 points)

(d) **Prove** that the collection of Turing decidable languages is closed under intersection. (2 points)

Problem 6 (Bonus): Turing Machines (5 points)

Let $HALT_{TM} = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on input } w\}$. Given that $HALT_{TM}$ is undecidable, **prove** that the collection of Turing recognizable languages is not closed under complement.

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.

There are six problems in total. You must solve the first five, and problem 6 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^n 1^m : n = 3q + r_1, m = 3q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 2\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ is a palindrome}\}$$
$$L_2 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : \text{every second letter in } w \text{ is } \mathtt{b}\}$$
$$L_3 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ contains more a than b}\}$$
$$L_4 = L_1 \cap L_2$$
$$L_5 = L_2 \cap L_3$$

Now solve the following problems.

(a) **Give** a context-free grammar that generates $\overline{L}_1$. (3 points) [Recall: $\overline{L}_1$ denotes the complement of the language L_1 i.e., $\overline{L}_1 = \Sigma^* - L_1$]

(b) **Give** a context-free grammar that generates L_2. (3 points)

(c) **Give** a context-free grammar that generates L_3. (3 points)

(d) **Write** all five-letter strings in L_4. (1 point)

(e) **Give** a context-free grammar that generates L_4. (2 points)

(f) **Give** a context-free grammar that generates L_5. (2 points)

(g) **Find** the total number of strings that belong to $L_5 \setminus L_1$. (1 point) [Recall: $L_5 \setminus L_1$ contains all strings that are in L_5 but not in L_1.]

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \to P0Q$$
$$P \to 0P0 \mid 0P1 \mid 1P0 \mid 1P1 \mid 0 \mid 1$$
$$Q \to 0Q1 \mid 1Q0 \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string '1100101010'. (3 points)

(b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)

(d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ starts with } 0 \text{ and ends with } 1\}$$

$$L_2 = \{w = 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } k = 2i + j\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and the number of 0s in } w_1 \text{ is equal to the number of 1s in } w_2\}$$

Now solve the following problems.

(a) **Give** the state diagram of a PDA that recognizes L_1. (3 points)

(b) **Give** the state diagram of a PDA that recognizes L_2. (3 points)

(c) **Give** the state diagram of a PDA that recognizes L_3. (4 points)

Problem 5 (CO4): Turing Machines (10 points)

Consider the following languages.

$$L_1 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and } \text{count}(w_1, 0) = \text{count}(w_2, 0)\}$$

$$L_2 = \{w \# w' : w \in \{0,1\}^*\}$$

(a) For the configuration $011q_4010$ of a Turing Machine, **what** is the current position of the tape head? (1 point)

(b) **Give** the state diagram of a turing machine that decides L_1. [Note: $\text{count}(w, a)$ denotes the number of occurrences of the symbol a in the string w.] (3 points)

(c) **Give** the state diagram of a turing machine that decides L_2. [Note: w' denotes the bitwise complement of w (obtained by flipping each bit: $0 \mapsto 1$, $1 \mapsto 0$), for example, if $w = 01101$, then $w' = 10010$.] (4 points)

(d) **Prove** that the collection of Turing decidable languages is closed under intersection. (2 points)

Problem 6 (Bonus): Turing Machines (5 points)

Let $HALT_{TM} = \{\langle M, w \rangle : M \text{ is a TM and } M \text{ halts on input } w\}$. Given that $HALT_{TM}$ is undecidable, **prove** that the collection of Turing recognizable languages is not closed under complement.

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.

FINAL EXAM
TOTAL MARKS: 50
DURATION: 100 MINUTES

A

There are a total of five problems. You have to solve all of them.

Problem 1 (CO3): Designing Context-Free Grammars (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following pair of languages. Recall that for a string w, $|w|$ denotes the length of w.

$$L_1 = \{w \in \Sigma^* : w \text{ contains exactly two } 1\text{s}\}$$
$$L_2 = \{x\#y : x \in \Sigma^*, y \in L_1, |x| = |y|\}$$

Now solve the following problems.

(a) **Give** a context-free grammar for the language L_1. (4 points)

(b) **Write** down four seven-letter strings in L_2. (1 point)

(c) **Give** a context-free grammar for the language L_2. (5 points)

Problem 2 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$S \to AB \mid \mathtt{b}AA \mid SS$$
$$A \to \mathtt{a} \mid \mathtt{a}A$$
$$B \to \mathtt{b} \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string abaaab. (3 points)

(b) **Sketch** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Demonstrate** that the given grammar is ambiguous by showing two more parse tree (apart from the one you already found in (b)) for the same string. (4 points)

(d) **Find** a string w of length five such that w has exactly one parse tree in the grammar above. (1 point)

Problem 3 (CO4): The CYK Algorithm (10 points)

Apply the CYK algorithm to determine whether the string yxzyz can be derived in the following grammar. You must show the entire CYK table. Here x, y and z are terminals, and the rest are variables.

$$S \to AB$$
$$A \to AC \mid \mathrm{y}$$
$$B \to BD \mid EB \mid \mathrm{z}$$
$$C \to AE \mid \mathrm{x}$$
$$D \to \mathrm{y}$$
$$E \to BD \mid \mathrm{x}$$

Problem 4 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

(a) **List** the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (4 points)

$$S \to \mathtt{a} \mid AB$$
$$A \to SB \mid \mathtt{ba} \mid B$$
$$B \to b \mid \varepsilon$$

(b) **Write** down the additional rules that need to be added to the following grammar if the production $X \to \varepsilon$ is removed. (4 points)

$$S \to \mathtt{a}X\mathtt{b}X\mathtt{c}X \mid X$$
$$X \to \mathtt{a} \mid \varepsilon$$

(c) **Write** down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$S \to A\mathtt{a}$$
$$A \to BC \mid C$$
$$B \to \mathtt{b}$$
$$C \to A \mid AA \mid A\mathtt{a}B$$

Problem 5 (CO3): Constructing Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Note that we define $c(w, x)$ to be the count of x in the string w.

$$L_1 = \{w \mid \ w \text{ is a palindrome and the length of } w \text{ is odd.}\}$$
$$L_2 = \{w_1\#w_2\#w_3 \mid \ c(w_1, 0) = c(w_2, 0) \text{ or } c(w_2, 1) = c(w_3, 1)\}$$

Now solve the following problems.

(a) **Give** the state diagram of a pushdown automaton that recognizes L_1. (4 points)

(b) **Find** all strings $w \in L_2$ such that w starts with `00010#1001#` and has a length of 14. (1 point)

(c) **Give** the state diagram of a pushdown automaton that recognizes L_2. (5 points)